

04 — Effort Estimation

All numbers in this document trace back to `params.md` (v2). When a figure says "per params," it means the Foundation Parameters document in this same directory.

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1. Executive Summary

The Health Hub v2 effort model provides a comprehensive, ground-up estimation of the total effort required to build and safely launch the platform. The model establishes a total effort baseline of **2,195 hours**, incorporating a direct team estimate for the remaining web platform work along with fixed Android API commitments, QA/DevOps hardening, and parallel operational workstreams.

A common mistake is to add only the two build workstreams (745 + 480 = 1,225 hours) and declare "that is all the work." It is not. The company is not shipping a side project. It is building a healthcare operation that requires training protocols, callback SOPs, partner onboarding kits, compliance documentation, launch analytics, and release governance. Those workstreams consume real hours from the same part-time team and are modeled explicitly.

Combined Effort Basis

Workstream Category	Hours
Web platform hardening	745
Android APK	480
QA, DevOps, security hardening	220
Parallel operational workstreams	750
Total	2,195

This is the number the rest of the business plan builds timelines and costs against.

Key Changes from v1

The v1 Problem: The v1 effort model used a three-point estimate (optimistic / expected / pessimistic) for remaining web work, producing a range of 682-1,413 hours. That range was too wide to plan against, and the expected-case value overweighted the web platform relative to actual team assessment.

What Changed: The development team provided a direct estimate: remaining web work is **150-160% of the Android effort** (per params.md, Section 6.1). With Android fixed at 480 hours, this yields a web estimate of 720-768 hours, modeled at **745 hours** (the 1.55x midpoint, rounded).

2. Revised Effort Stack

2.1 Full Workstream Table

#	Workstream	Hours	Rate	Cost	Rationale
1	Web platform hardening	745	\$20/hr (senior)	\$14,900	480 hrs Android x 1.55 multiplier = 744, rounded to 745. Covers security hardening, role-flow completion (GP, specialist, pharmacy, diagnostics, admin), SSR safety, auth hardening, rate limiting, API hardening, error handling, monitoring integration, performance optimization, and load testing. Per params.md Section 6.
2	Android APK	480	\$15/hr (blended)	\$7,200	Fixed external quote, already contracted. Patient-facing mobile application covering registration, dashboard, consultation, video/audio/chat, prescriptions, lab results, notifications, payments. Per params.md Section 5.
3	QA, DevOps, security hardening	220	\$20/hr (senior)	\$4,400	Release discipline, CI/CD pipeline maturity, monitoring stack, penetration testing, security audit remediation, load testing infrastructure, rollback procedures. Not included in the web or Android line items — this is cross-cutting infrastructure work.
4	Doctor/admin training and SOP design	180	\$15/hr (blended)	\$2,700	Doctor training protocols and playbooks (120-160 hrs per params.md Section 17.1, midpoint 140) + admin callback SOPs and escalation trees (80-120 hrs, midpoint 100). Combined and de-duplicated to 180 hrs because training and SOPs share content overlap.
5	Provider onboarding and partner tooling	140	\$15/hr (blended)	\$2,100	Partner onboarding kits for clinics, labs, pharmacies (80-120 hrs per params.md Section 17.1, midpoint 100) + staffing calibration and refresher training (40-60 hrs, midpoint 50). Combined to 140 hrs accounting for overlap.
6	Launch analytics and reporting	120	\$15/hr (blended)	\$1,800	Launch analytics and reporting setup (60-80 hrs per params.md, midpoint 70) + release governance and rollback checklists (40-60 hrs, midpoint 50). Combined to 120 hrs.
7	Admin operational guidelines	80	\$15/hr (blended)	\$1,200	Admin operational guidelines covering daily workflows (60-80 hrs per params.md Section 17.1). Modeled at 80 hrs (top of range) because these are live-ops documents that require iteration during pilots.

8	Compliance and regulatory documentation	100	\$15/hr (blended)	\$1,500	Compliance and regulatory documentation (80-120 hrs per params.md Section 17.1, midpoint 100). Covers Ethiopia MOH requirements, Kenya PPB oversight, data protection documentation, and telehealth regulatory filings.
9	Technical support runbooks	60	\$15/hr (blended)	\$900	Technical support runbooks (60-80 hrs per params.md Section 17.1). Modeled at 60 hrs (bottom of range) because much content derives from existing CLAUDE.md audit documentation and agent-state records.
10	Remaining parallel ops buffer	70	\$15/hr (blended)	\$1,050	Residual operational effort to reach the 750-hour parallel ops total. Covers ad hoc founder time on investor materials, legal coordination, market research updates, and unforeseen operational setup tasks.
	Total	2,195		\$37,750	

2.2 Workstream Grouping Summary

Group	Workstreams	Hours	% of Total
Pure build (software)	#1 + #2	1,225	55.8%
Cross-cutting technical	#3	220	10.0%
Operational setup	#4 + #5 + #6 + #7 + #8 + #9 + #10	750	34.2%
Total	All	2,195	100%

2.3 Cost Reconciliation

Segment	Hours	Rate Basis	Cost
Web platform hardening	745	\$20/hr (senior dev rate per params.md Section 3.1)	\$14,900
Android APK	480	\$15/hr (blended rate per params.md Section 5)	\$7,200
QA/DevOps/security	220	\$20/hr (senior dev rate — DevOps + Technical Head)	\$4,400
Parallel ops (all)	750	\$15/hr (blended — mix of founder, founding team, advisors)	\$11,250
Total	2,195		\$37,750

This matches the combined effort summary in params.md Section 17.3.

3. Technical vs Non-Technical Split

3.1 The Split

Category	What It Includes	Hours	Cost
Pure technical build and hardening	Web hardening (745) + Android APK (480) + QA/DevOps/security (220)	1,445	\$26,500
Operations setup and launch governance	Training (180) + partner onboarding (140) + analytics/reporting (120) + admin ops guidelines (80) + compliance (100) + support runbooks (60) + ops buffer (70)	750	\$11,250
Total		2,195	\$37,750

3.2 Why the Split Matters

The 1,445 technical hours are what most software project estimates would include. The 750 operational hours are what most software project estimates would miss.

Health Hub is not a consumer app that ships and self-serves. It is an admin-assisted healthcare platform where:

- Doctors must be trained on the platform's consultation workflow before they see patients.
- Admin staff must follow documented callback SOPs or service quality collapses.
- Partner clinics, labs, and pharmacies need onboarding materials or they will not adopt the platform.
- Regulatory compliance documentation is a legal prerequisite in both Ethiopia and Kenya.
- Launch analytics must exist before Pilot 1 or the company flies blind through its first external test.

Excluding these workstreams produces a plan that is technically complete but operationally unlaunchable.

3.3 Who Does What

Category	Primary Contributors	Per Params.md
Web hardening	2 web developers (DevOps + Full-stack) + 1 flex developer	Section 2.2
Android APK	3-4 Android/backend developers	Section 2.2
QA/DevOps/security	Technical Head + DevOps engineer	Section 2.1, 2.2
Training and SOPs	Founding Member 1 + Medical Advisor	Section 17.1
Partner onboarding	Owner + Founding Member 1	Section 17.1
Analytics and governance	Technical Head	Section 17.1
Compliance and regulatory	Owner + Advisors	Section 17.1

4. Team Capacity Model

4.1 Raw Capacity Math

Per params.md Section 2, the team consists of 12 part-time contributors, all working approximately 4 hours/day, 22 working days/month = ~88 hours/month each.

Group	People	Hours/Person/Month	Raw Group Capacity
Core team	3	88	264 hrs/mo
Development team	7	88	616 hrs/mo
Advisory board	2	25 (avg)	50 hrs/mo
Total	12		930 hrs/mo

4.2 Why Raw Hours Are Not Productive Hours

The 930 hrs/mo raw capacity is a theoretical ceiling. Real output is reduced by:

Overhead Factor	Impact	Who It Affects
Coordination and handoff	10-15% loss	Everyone — part-time schedules mean frequent context switches and async handoffs
Testing and rework	10-15% loss	Developers — bugs, failed builds, regression fixes
Product decisions and review	5-10% loss	Core team + senior devs — design reviews, scope decisions, PR reviews
Partner and ops interruptions	5-10% loss (grows after Pilot 1)	Core team — investor calls, partner meetings, live issue triage
Live issue handling	0% (Pre-Pilot) to 15% (Pilot 2+)	Grows as real users create real support load
Role-switching cost	5-10% loss	Founders who split time between build, ops, and business development

Combined overhead: 30-50% of raw hours, depending on phase and investment level.

4.3 Productive Capacity by Investment Level

Level	Label	Productive Hrs/Mo	Derivation
L1	Bootstrapped	~320 hrs/mo	~35% of 930 raw. Maximum constraint: more sequencing, founder time heavily split, lean utilization. Per params.md Section 2.4.
L2	Ideal	~480 hrs/mo	~52% of 930 raw. Balanced baseline: enough concurrency for parallel build + operating prep. Per params.md Section 2.4.
L3	Fully Funded	~640 hrs/mo	~69% of 930 raw. Full effective utilization: proper tooling, less context switching, dedicated roles. Per params.md Section 2.4.

4.4 Months to Complete at Each Level

Using the total effort of 2,195 hours:

Level	Productive Hrs/Mo	Months (2,195 / capacity)	Calendar Interpretation
L1	320	6.9 months of pure work	~8-10 calendar months (sequencing adds overhead)
L2	480	4.6 months of pure work	~6-8 calendar months (balanced parallelism)
L3	640	3.4 months of pure work	~5-6 calendar months (true parallelism, but floor exists)

Important caveat: These are effort-to-capacity ratios, not phase-by-phase timelines. The actual calendar schedule is longer because:

1. Not all work can be parallelized (dependencies exist between workstreams).
2. Pilot phases require elapsed calendar time for user feedback loops.
3. Operational workstreams (training, SOPs, compliance) cannot be front-loaded entirely — they require iteration during live pilots.
4. Staffing ramp-up for operational employees happens during pilots, not before.

The phase-by-phase timeline is modeled in the timeline document (05-timelines.md), not here. This document establishes the effort inputs.

4.5 Capacity Split: Technical vs Operational

Not all productive hours go to code. The capacity model must serve both the technical and operational workstreams simultaneously.

Level	Total Productive	Technical Allocation	Ops Allocation	Reasoning
L1	320 hrs/mo	~220 hrs/mo	~100 hrs/mo	At L1, ops is squeezed — founders handle both; sequencing required
L2	480 hrs/mo	~320 hrs/mo	~160 hrs/mo	Balanced: dev team runs build, core team runs ops with some overlap
L3	640 hrs/mo	~420 hrs/mo	~220 hrs/mo	Full utilization: dedicated ops effort, developers unblocked

5. Effort by Phase

5.1 Phase Effort Distribution

The 2,195 total hours are not consumed evenly. Each phase has a different effort profile based on what work is possible and necessary at that stage.

Phase	Duration Range	Technical Hrs	Ops Hrs	Total Hrs	% of Total	Primary Character
Pre-Pilot	3-6 months	650-750	150-200	800-950	~40%	Heaviest technical concentration
Pilot 1	2-3 months	300-350	180-220	480-570	~24%	Mixed technical and operating load
Pilot 2	3-13 months	250-300	200-250	450-550	~23%	Scale-up and operational maturity
Production Readiness	4-12 months	100-150	170-220	270-370	~13%	Hardening and governance
Total		~1,445	~750	~2,195	100%	

5.2 Pre-Pilot Effort Detail (~40% of total)

This is the most engineering-intensive phase. The team must convert the prototype into something safe enough for external users.

Workstream	Hours in Phase	Key Deliverables
Web hardening	350-400	Architecture freeze, auth hardening, SSR safety, role-flow completion, API hardening, rate limiting
Android APK	200-250	Core patient journeys: registration, dashboard, consultation request, basic video/audio/chat
QA/DevOps	100-120	CI/CD pipeline, monitoring stack, staging environment, initial load testing
Training and SOPs	80-100	Draft doctor training protocols, initial admin callback SOPs, first-pass operational guidelines
Partner onboarding	40-60	Onboarding kit drafts for first pilot clinics, pharmacy partner agreements
Compliance	30-40	Ethiopia regulatory research, initial telehealth compliance documentation

5.3 Pilot 1 Effort Detail (~24% of total)

The character of work shifts. Live users create support obligations. Bugs surface that internal testing missed. The team must fix and build simultaneously.

Workstream	Hours in Phase	Key Deliverables
Web hardening	150-180	Live bug fixes, performance optimization under real load, feature completion for active role flows
Android APK	150-180	Full scope completion: notifications, payments, prescription viewing, lab results
QA/DevOps	60-70	Production monitoring, incident response procedures, security patching
Training and SOPs	60-70	Doctor refresher training based on real consult data, admin SOP refinement from live callbacks
Support runbooks	30-40	L1/L2 support documentation based on actual ticket patterns
Analytics	30-40	Initial reporting dashboards, user activity tracking, service quality metrics

5.4 Pilot 2 Effort Detail (~23% of total)

Scale-up creates new problems. The platform serves 5,000-10,000 users. Staffing ladder activates. Partner network expands.

Workstream	Hours in Phase	Key Deliverables
Web hardening	150-180	Scalability hardening, advanced reporting, partner admin tooling
Android APK	80-100	Polish, edge-case fixes, performance optimization for diverse Android devices
QA/DevOps	40-50	Load testing at scale, automated regression, security re-audit
Partner onboarding	60-80	Expanded partner onboarding (Kenya prep), lab and pharmacy network growth
Staffing calibration	40-60	Refresher training as operational staff onboard, shift pattern optimization
Analytics and reporting	50-60	Investor-quality reporting, unit economics tracking, LTV:CAC measurement
Compliance	40-50	Kenya regulatory documentation, cross-border data handling policies

5.5 Production Readiness Effort Detail (~13% of total)

The least feature-heavy phase. The focus is on making what exists safe, governed, and sustainable.

Workstream	Hours in Phase	Key Deliverables
Web hardening	50-80	Final security hardening, penetration test remediation, release freeze
QA/DevOps	30-40	Production release checklists, rollback procedures, disaster recovery testing
Compliance	30-40	Final regulatory submissions, data protection audit, compliance sign-off
Launch governance	40-50	Launch checklists, support schedules, escalation trees, on-call rotation
Support maturity	30-40	Full support runbook completion, L1/L2/L3 escalation documentation
Ops buffer	30-40	Final operational setup, staff training completion, go-live preparation

6. Why Timelines Did Not Collapse

6.1 The Core Question

If v2 reduced web effort from the v1 expected case of ~1,050 hours to 745 hours (a ~30% reduction), why didn't the overall timeline shrink by 30%?

6.2 The Answer in Three Parts

Part 1: The web estimate went down, but total work did not shrink proportionally.

Component	v1 Model	v2 Model	Change
Web effort	~1,050 hrs (expected)	745 hrs	-305 hrs (-29%)
Android effort	440-640 hrs (range)	480 hrs (fixed)	Clarified, not reduced
QA/DevOps/security	Implicit / partially hidden	220 hrs (explicit)	Now visible
Parallel ops	Not modeled	750 hrs (explicit)	Now visible
Total	~1,885 (reconstructed)	2,195	+310 hrs (+16%)

The v2 total is actually *higher* than v1 because v1 did not model operational workstreams. The web estimate came down, but the honest total went up.

Part 2: The company still needs to build non-software systems.

Even if coding finished tomorrow, Health Hub could not launch because:

- No doctor has been trained on the platform's consultation workflow.
- No admin has a documented callback SOP to follow.
- No partner clinic has an onboarding kit.
- No compliance documentation exists for Ethiopia or Kenya regulators.
- No launch analytics are in place to measure whether the pilot is working.
- No support runbook exists for the technical support staff the company must hire.

Each of these workstreams requires elapsed calendar time, iteration during pilots, and real feedback loops. They cannot be parallelized away.

Part 3: Timeline compression has a floor.

Even at L3 (full utilization, 640 hrs/mo), the minimum calendar time is bounded by:

Constraint	Minimum Elapsed Time	Why
Pilot 1 feedback loop	2-3 months	Real users need time to use the platform and generate actionable data
Staff hiring and training	1-2 months	Operational employees must be recruited, hired, and trained before they are effective
Regulatory submissions	1-3 months	Government agencies have their own processing timelines
Partner activation	1-2 months	Clinics, pharmacies, and labs have their own decision and integration cycles
Android APK completion	2-4 months	480 hours at 3-4 developers is not instant, especially with backend dependencies

These constraints create a timeline floor that pure coding speed cannot break through.

6.3 What Did Compress

The timeline did get shorter in v2 — just not by 30%. The compression shows up as:

- Shorter Pre-Pilot phase (less web work to complete before first external users).
- Earlier Pilot 1 entry (product reaches minimum viable state sooner).
- More founder time freed for operational workstreams during build phase.
- Higher confidence in the web estimate (narrower range = less contingency padding).

7. Interpretation by Investment Level

7.1 Level 1 — Bootstrapped (~320 hrs/mo)

Dimension	L1 Experience			
Sequencing	Heavy. Android and web largely sequential. Ops work fits into gaps.			
Founder load	Maximum. Founders split between build oversight, ops, partner work, and investor relations.			
Phase duration	Longest. Pre-Pilot stretches to 5-6 months. Pilot 1 to 3 months. Total 22-34 months.			
Risk	Highest. Slow pace means competitors or market shifts can erode first-mover advantage.			
Effort math	2,195 hrs / 320 hrs/mo = 6.9 months pure work. Calendar: ~10+ months with sequencing.			

L1 effort allocation by month (illustrative):

Month	Technical	Ops	Total	Notes
1-2	200	60	260	Web hardening sprint, Android kickoff, initial SOP drafting
3-4	220	80	300	Android ramp-up, auth/security hardening, training protocol first draft
5-6	200	100	300	Feature completion push, partner onboarding starts, compliance research

7-8	180	120	300	Pilot 1 entry, live bug fixes, callback SOPs refined
9-10	160	140	300	Pilot 2 prep, analytics build, staffing calibration

7.2 Level 2 — Ideal (~480 hrs/mo)

Dimension	L2 Experience
Sequencing	Moderate. Android and web run in parallel. Ops work has dedicated time.
Founder load	Balanced. Founders can focus on ops and business while dev team handles build.
Phase duration	Moderate. Pre-Pilot at 3-4 months. Pilot 1 at 2-3 months. Total 16-24 months.
Risk	Moderate. Pace is fast enough to maintain momentum without burning the team.
Effort math	2,195 hrs / 480 hrs/mo = 4.6 months pure work. Calendar: ~7-8 months with dependencies.

L2 is the baseline planning case. The effort-to-outcome ratio is most favorable here because the team can sustain concurrency without the coordination cost exploding.

7.3 Level 3 — Fully Funded (~640 hrs/mo)

Dimension	L3 Experience
Sequencing	Minimal. True parallelism across Android, web, and ops.
Founder load	Lightest. Founders can focus on strategy, investors, and market while team executes.
Phase duration	Shortest. Pre-Pilot at 3 months. Pilot 1 at 2 months. Total 12-20 months.
Risk	Lowest technical risk, but highest coordination cost. More people = more meetings.
Effort math	2,195 hrs / 640 hrs/mo = 3.4 months pure work. Calendar: ~5-6 months with floor constraints.

L3 caveat: Even at maximum capacity, the timeline cannot compress below the floor constraints identified in Section 6.2. The extra capacity shows up as *better quality* and *earlier operational readiness* rather than a proportionally shorter calendar.

7.4 Comparative Summary

Metric	L1	L2	L3
Productive hrs/mo	320	480	640
Pure effort months	6.9	4.6	3.4
Realistic calendar months (to Pilot 1)	7-9	5-6	4-5
Realistic calendar months (to Production)	22-34	16-24	12-20
Monthly team cost (per params.md)	~\$14,830	~\$14,830	~\$14,830
Cumulative team cost to Production	\$326K-504K	\$238K-356K	\$178K-297K
Best-fit scenario codes	S1-L1, S2-O1-lx	S2-O2-lx, S3-O2-lx	S1-L3, S2-O3-lx, S3-Ox-l3

Note on cost: Monthly team cost is the same at all levels because the team is fixed at 12 part-time people. The difference is *utilization rate*, not headcount. L3 does not add people — it extracts more productive hours per person through better tooling, fewer context switches, and dedicated role focus. The total cost difference comes from timeline duration: shorter timelines mean fewer months of burn.

8. Closing Note

The v2 effort model corrects two problems from v1:

- Web estimate accuracy.** The web effort is now anchored to a real team estimate (150-160% of Android) rather than a wide three-point spread. This produces a tighter, more plannable number.
- Operational effort visibility.** The 750 hours of parallel operational work are now explicitly modeled, line-itemized, and phase-allocated. This prevents the plan from appearing artificially optimistic about timelines while also making the non-engineering investment legible to investors.

The combined 2,195-hour model is the foundation for the timeline, cost, and scenario documents that follow.

All figures trace to params.md v2. For rate card details, see params.md Section 3. For phase definitions, see params.md Section 13. For scenario codes, see params.md Section 14.